

Cross-Layer Observability for Intrusion Detection in Confidential AI Training Clusters

A. Barreiro^(*,1), J. G. Meyer^(1,2), D. C. Lawo^(1,2), I. Tafur Monroy⁽¹⁾, J. J. Vegas Olmos⁽²⁾

(1) Electrical Engineering Department, Eindhoven University of Technology (TU/e), Eindhoven, The Netherlands

(2) Networking SW & Sys Arch, NVIDIA Corporation, Yokneam Illit, Israel

*a.barreiro@tue.nl

Abstract—Modern AI training clusters increasingly pair confidential computing with end-to-end encryption on east–west Remote Direct Memory Access links, making payload-level deep packet inspection infeasible and constraining host-based intrusion detection systems through kernel bypass and enclave isolation. Addressing this gap, this paper presents a novel systematic intrusion-detection framework for this setting. We map thirteen attacks directly into three detectability tiers based on their visibility across a three-layer signal surface (network, infrastructure, and workload) that survives encryption and trusted execution environment boundaries. Central to our framework is defensive inversion, a new detection strategy that repurposes side-channel-style signals as anomaly-detection features. We also provide the outline of a cross-layer DPU-based monitoring architecture that applies our framework to confidential training clusters.

Index Terms—confidential computing, intrusion detection, RDMA, distributed training, DPU telemetry

I. INTRODUCTION

Large-scale AI training clusters are increasingly adopting confidential computing to protect model weights, training data, and gradients within hardware-enforced Trusted Execution Environments (TEEs). At the same time, operators are moving toward end-to-end encryption for east–west traffic, including Remote Direct Memory Access (RDMA) fabrics, typically via IPsec or MACsec. Together, these trends undermine a core assumption of traditional intrusion detection: that a trusted monitor can directly observe the system being monitored [1]. In particular, they restrict (i) *payload visibility* on the network, since encrypted RDMA prevents deep packet inspection, and (ii) *host-level introspection*, since RDMA kernel bypass removes the OS from the data path while TEE isolation is designed to prevent infrastructure operators from observing enclave computation. As a result, intrusion detection in confidential training clusters must shift from payload inspection to inference over the limited signals that remain observable.

Encrypted traffic analytics and metadata-only detectors, e.g., [2] and [3], provide a partial solution to the lack of payload visibility. However, these methods were developed around TCP/IP enterprise or general data-center traffic and do not match the traffic dynamics of distributed training. In modern GPU clusters, collective communication (e.g., AllReduce) induces highly periodic, bursty, synchronization-driven patterns whose features and model assumptions differ substantially from conventional flow-based settings.

Meanwhile, the community has grown increasingly aware of RDMA-specific threats [4], [5], vulnerabilities in distributed training [6], and the security requirements of confidential AI infrastructure [7]. However, no current intrusion detection system (IDS) is tailored to the intersection of RDMA-based GPU communication, collective synchronization dynamics, and confidential computing. One natural direction is privacy-preserving IDS (e.g., anomaly detection on encrypted features via homomorphic encryption), which reduces plaintext exposure but imposes steep performance and feature constraints and does not target line-rate RDMA collective behavior [8]–[11]. In the opposite direction, recent cross-layer timing analyses such as [12] and [13] provide strong observability of distributed training behavior, but are oriented toward reliability rather than adversarial threats and typically do not assume strict encryption or semi-trusted-operator constraints.

Building on these gaps, we present a systematic framework for intrusion detection in confidential AI training clusters, where both payload visibility and host-level introspection are restricted by design. Our goal is to make explicit what can still be observed under encryption and TEEs, which attacks remain distinguishable under this observation model, and how to design detectors that rely only on the surviving signals.

We structure the paper as follows.

- In Sec. II, we formalize confidential AI intrusion detection as a distinct problem setting, classifying thirteen relevant attacks directly into three detectability tiers based on whether they produce observable signals under encryption and TEEs.
- In Sec. III, we map these attacks across a three-layer signal surface, identifying the specific metadata features that survive these constraints. As part of this, we introduce *defensive inversion*: the practice of repurposing signals traditionally treated as side-channel vulnerabilities as defensive anomaly-detection features.
- In Sec. IV, we propose a cross-layer DPU-based monitoring architecture that aligns with these three signal layers to detect the identified threats.
- Finally, Sec. V is devoted to discussions and concluding remarks.

II. ADVERSARY MODEL AND ATTACK CLASSIFICATION

We consider an adversary that may sit *anywhere outside the TEE boundary*. This includes a compromised or malicious

worker node deviating from the training protocol, a privileged infrastructure component such as a rogue administrator or hypervisor, and a co-located tenant sharing physical links. Confidential computing and end-to-end encryption prevent such an adversary from reading or modifying payloads in transit, or extracting plaintext data from TEE memory. They *can*, however, observe traffic metadata (e.g. packet timing, flow sizes, and RDMA congestion signals) and manipulate traffic timing, volume, or routing.

We then identify 13 specific attacks and classify them into three distinct detectability tiers based on what can and cannot be observed when payloads are encrypted. This classification is summarized in Table I and is based on known frameworks against RDMA fabrics [4], [5], Byzantine fault tolerance literature [6], [14], and gradient poisoning research [15], [16].

Tier 1 (Strongly Detectable): Identifiable through metadata and timing signals alone. Even without payload visibility, attacks in this tier produce robust, encryption-invariant anomalies in network flow statistics, hardware performance counters, or collective synchronization timing.

Tier 2 (Partially Detectable): Detectable only if specialized architectural monitoring is in place. Attacks in this tier attempt to bypass protections via side channels or subtle workload manipulations, requiring continuous introspection into specific hardware telemetry or framework-exposed statistics to detect.

Tier 3 (Fundamentally Undetectable): Undetectable by any metadata-only IDS. Attacks in this tier operate entirely within the encrypted payload or passively observe the system without perturbing timing, message sizes, or hardware counters. Without inspecting payload content or running redundant structural checks, these exploits are identical to benign behavior, forming an *information-theoretic* wall.

The IDS framework that we propose in this paper explicitly targets integrity and availability threats (Tier 1 and 2). Passive confidentiality attacks remain outside our detection perimeter, an unavoidable reality for any infrastructure-level IDS operating in fully encrypted environments.

III. CROSS-LAYER OBSERVABLE SIGNALS

We classify the observable signal surface into three layers: network, infrastructure, and workload, mapping each to detectable attacks (Tiers 1 and 2) from Sec. II. This classification is based on RDMA congestion control research [17], [18],

GPU interconnect side-channel studies [19], [20], and recent AI cluster observability work [12].

A. Network-Layer Signals

Strict end-to-end encryption protects payloads, not headers or timing. As a result, the network fabric exposes rich metadata, and flow-level observables such as packet inter-arrival times, flow sizes, completion times, and fan-out topology remain fully intact. These flow-level signals separate normal AllReduce traffic, which appears as highly periodic bursts with symmetric sizes, from anomalies such as asymmetric flooding or bandwidth squatting.

In addition, RDMA congestion signals embedded in the RoCEv2 protocol stack provide a distinct fingerprint. Under DCQCN [17], three signals interact in a tightly coupled sequence: switch-level congestion triggers Explicit Congestion Notification (ECN) marks, receivers respond with Congestion Notification Packets (CNPs), and saturated loops trigger Priority Flow Control (PFC) pauses. An adversarial congestion attack perturbs all three signals simultaneously. These, along with RDMA retransmission counters, live on the NIC ASIC and are directly readable by the DPU.

B. Infrastructure-Layer Signals

The second layer of observability comes from the hardware telemetry produced by RDMA NICs and GPU interconnects. On the one hand, *RDMA hardware counters* track per-Queue-Pair (QP) state, completion queue (CQ) overflows, and protection domain violations. Polling these counters from the BlueField DPU using the DOCA SDK [4] consumes minimal overhead. Counters like `cq_overrun` and `local_protection_err` should be strictly zero during healthy training; any deviation is a clear anomaly.

On the other hand, *NVLink performance counters* track per-link TX/RX byte counts, throughput, and utilization for GPUs communicating within a node. Although previous work [19], [20] viewed the timing of NVLink contention exclusively as an attack vector for leaking information, we propose *defensive inversion*, repurposing these signals as defensive anomaly-detection features. By monitoring NVLink TX/RX symmetry, we can detect data staging; by tracking contention timing, we can spot unauthorized workloads (rogue nodes); and by comparing observed traffic against expected AllReduce patterns, we can verify workload integrity.

TABLE I: Attack classification by detectability tier.

Tier 1 Strongly Detectable	Tier 2 Partially detectable	Tier 3 Fundamentally Undetectable
RDMA flooding / resource exhaustion	TEE side-channel attacks	Training data poisoning
Bandwidth squatting / congestion inj.	Gradient poisoning (norm-infl.)	Memory bus snooping
BGP / routing manipulation		Passive timing observation
Byzantine node slowdown		Covert timing channels
AllReduce / collective interference		Model / IP theft via side channels
Rogue GPU node behavior		

TABLE II: Signal-attack matrix for Tier 1 and Tier 2 attacks.

Symbols: $\uparrow\uparrow$ strong perturbation, $\uparrow$ moderate, $\sim$ weak, $—$ none. Layers: (N) network, (I) infrastructure, (W) workload.

Observable Signal	<i>DPU-obs.</i>	<i>RDMA Flood</i>	<i>BW Squatting</i>	<i>Routing Manip.</i>	<i>Byzantine Slow</i>	<i>AllReduce Interf.</i>	<i>Rogue Node</i>	<i>TEE Side-Ch.</i>	<i>Grad. Poison.</i>
Tier 1					Tier 2				
Packet timing / jitter (N)	✓	⇕	⇕	⇕	↑	⇕	↑	—	—
Flow size / rate (N)	✓	⇕	↑	—	—	—	↑	—	~
RDMA retx / CNP (N)	✓	⇕	↑	↑	—	↑	~	—	—
ECN / PFC counters (N)	✓	⇕	⇕	—	—	—	—	—	—
NIC queue depth (I)	✓	⇕	↑	—	—	—	—	—	—
Link utilization (I)	✓	⇕	⇕	↑	—	—	↑	—	—
NVLink TX/RX symmetry (I)	✓	~	~	—	—	—	⇕	—	—
HW perf. counters (I)	—	—	—	—	—	—	—	⇕	—
GPU utilization (W)	—	~	~	—	⇕	↑	⇕	—	—
AllReduce timing (W)	—	↑	↑	↑	⇕	⇕	↑	—	—
Step duration (W)	—	↑	↑	↑	⇕	⇕	↑	—	~
Gradient norms (W)	—	—	—	—	—	—	—	—	⇕

C. Workload-Layer Signals

At the workload layer, synchronous distributed training produces unique behavioral signatures. AllReduce operations create a strict global synchronization barrier at every step, resulting in a highly periodic timing pattern [12]. By observing the latency of each rank’s contributions via NCCL¹ traces, we can spot straggler ranks that indicate Byzantine slowdowns or collective interference.

Additionally, because all ranks execute the exact same model on different data batches, GPU utilization should be nearly symmetric. Persistent deviations detected via per-second NVIDIA DCGM² indicate unauthorized workloads. Finally, runtime deviations in the static NCCL collective communication pattern signal node substitution or routing manipulation.

D. What Remains Blind

As established in Sec. II, Tier 3 attacks remain strictly opaque. Encryption and TEEs hide the payload content (weights, training data, gradient values), specific RDMA op-codes (encrypted inside RoCEv2), and the exact direction of the gradient. Detecting malicious shifts in the gradient direction without payload access is information-theoretically impossible [15], [16]. Our IDS operates exclusively on the metadata signals described in Sec. III-A, III-B, and III-C, accepting the blind spots in Tier 3 as hard constraints. Table II condenses the signal-attack mapping.

IV. DETECTION ARCHITECTURE

To process the various signals into actionable detection decisions, we propose a DPU-based architecture that directly monitors the three signal layers of interest. Our proposal

consists of a pipeline with five stages: *Collect*, *Extract*, *Score*, *Fuse*, and *Alert*, and builds on the metadata-only paradigm used by tools like NetVigil [3] and Kitsune [2], but customizes it specifically for RDMA traffic patterns and the strict constraints of confidential computing. Fig. 1 illustrates the proposed five-stage pipeline and its integration with the three monitored signal layers.

A. Collection

As illustrated in Fig. 2, the signal collection spans two vantage points that respect the observability limit of the TEE/network. Three telemetry streams feed the pipeline. *Flow metadata* and *NIC/NVLink counters* are both polled by the BlueField DPU, which runs its own OS on separate ARM cores outside the host’s TEE and reads NIC state directly via MMIO; a compromised host OS cannot tamper with or suppress either stream. Workload signals originate inside the TEE and are exported by an *attested host agent* that safely reports per-step NCCL profiler timings and per-second DCGM GPU metrics with negligible overhead [13]. All three streams are forwarded to a centralized detection engine, which can run on a dedicated management node or be distributed across multiple DPUs.

B. Per-Layer Feature Extraction and Scoring

Each layer uses a detector matched to its signal statistics. **Network layer (Threshold & Rate):** Attacks such as RDMA flooding or PFC storms cause significant spikes in congestion counters. For these threats, simple threshold and rate rules enable excellent low-latency detection, such as flagging when the ECN marking rate stays persistently above its workload-specific baseline, or when CNP rates and PFC pause durations spike [17]. The DCQCN protocol operates as a cascade, as explained in Sec. III-A, so an actual attack produces a temporally

¹NVIDIA Collective Communications Library.

²Data Center GPU Manager.

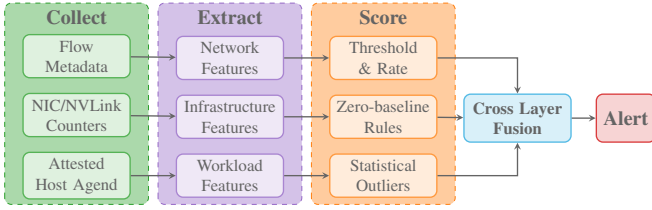


Fig. 1: Detection pipeline. Three telemetry streams (Collect) are converted to per-layer features (Extract) and scored by layer-matched detectors (Score). The fusion engine correlates the three scores across layers and triggers an attributed alert when joint conditions are met.

correlated signature across multiple counters, distinguishing it from organic congestion.

Infrastructure layer (Zero-baseline Rules): Zero-baseline RDMA counters, such as `cq_ouerrun` and `local_protection_err`, are unambiguous: any non-zero reading during steady-state training is an actionable anomaly. For NVLink metrics, we apply defensive inversion, continuously comparing observed TX/RX symmetry against the expected profile of the tensor-parallel topology; any *persistent* deviation is flagged as a potential workload-integrity violation.

Workload layer (Statistical Outliers): Under healthy conditions, AllReduce step timing follows a tight Gaussian distribution. By applying standard statistical outlier detection (e.g., Z-scores or median absolute deviation) to the per-rank collective duration, we can easily spot straggler ranks indicative of Byzantine slowdowns or collective interference. For low-rate attacks such as bandwidth squatting, we use cumulative sum (CUSUM) change-point detection to track mean-shift drift in step durations. Finally, if a node’s GPU utilization profile deviates from the cluster mean by more than two standard deviations, it is flagged as a potential rogue node.

C. Cross-Layer Fusion and Altering

As shown in Table II, no single signal layer covers all Tier 1 attacks. Detecting a rogue node, for example, requires combining anomalous network fan-out (network layer), GPU utilization deviation (workload layer), and NVLink asymmetry (infrastructure layer).

We adopt a *late-fusion* strategy: each per-layer score from Sec. IV-B is normalized to its own baseline, and the fusion engine flags joint conditions across the three scalars (e.g., an elevated network score concurrent with a workload timing deviation). The downstream *Alert* stage then attributes the detection to a specific attack type via the signal-attack correspondence in Table II. The detailed evaluation of the fusion engine and alert-attribution logic is the subject of ongoing work.

V. DISCUSSION AND CONCLUSION

In this paper, we reframe intrusion detection for confidential AI training clusters as an observability-constrained problem: encrypted RDMA removes payload visibility, and TEEs

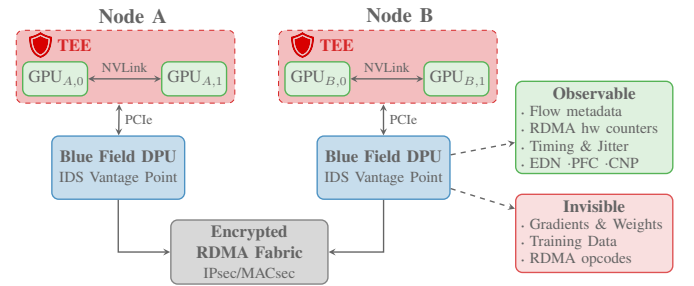


Fig. 2: Confidential AI training cluster architecture. Each GPU node executes within a TEE boundary (dashed). The BlueField DPU sits outside the TEE on the network path, serving as the IDS vantage point — it observes flow metadata and hardware counters while remaining blind to encrypted payloads and model data.

limit host introspection. We formalize this setting through an attacker model and organize potential threats into three detectability tiers based on which signals remain observable under these constraints. We then systematize the residual signal surface across network, infrastructure, and workload layers, introducing defensive inversion as a design principle to turn side-channel-like timing and contention artifacts into detection features. Finally, we propose a DPU-based monitoring architecture that collects, extracts, scores, and fuses these cross-layer signals into actionable alerts while preserving confidentiality.

Several open challenges remain. The first is attribution: our approach detects anomalous deviations, but cannot inherently distinguish deliberate attacks from benign faults. For example, a Byzantine slowdown can be indistinguishable from thermal throttling, and robust attribution remains an open problem. The second is non-stationary baselines, as training behavior shifts across phases such as warmup, steady-state, and convergence, requiring phase-aware profiling beyond the CUSUM-style per-layer detector in Sec. IV-B. The third is adaptive adversaries: a monitoring-aware attacker may operate just below detection thresholds. Low-rate attacks such as LoRDMA [5] can significantly degrade throughput while staying near the noise floor, which motivates the cross-layer fusion strategy we proposed in Sec. IV-C. Finally, scale remains a practical concern, as extending from small clusters to thousands of nodes will require hierarchical aggregation and DPU-side pre-filtering to keep telemetry volume and false-positive rates manageable.

As established in Sec. II, five Tier 3 attacks remain fundamentally out of reach for any metadata-only IDS. Most importantly, norm-correct gradient poisoning under encryption [15], [21] reflects an information-theoretic limitation rather than an engineering shortfall: when an attack produces no observable deviation in the available metadata, detection is impossible by design. Passive confidentiality attacks that leave no footprint are similarly beyond the scope. For these threats, the appropriate path is complementary defenses, e.g. encryption-compatible Byzantine fault tolerance such as Lagrange Coded

Computing [22], integrated alongside the metadata-based monitoring framework proposed here.

Future work includes hardware validation on a 2–4 node GPU cluster with ConnectX-7 NICs and BlueField-3 DPUs, running PyTorch Distributed Data Parallel with the NCCL backend. We will inject four Tier 1 attacks from Table I at multiple severity levels under both plaintext and IPsec-encrypted conditions, to verify that the proposed metadata features remain usable and encryption-invariant. Full experimental results will appear in a forthcoming paper.

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